1.2 - Row Reduction and Echelon Forms

Notes: The key exercises are 1–24 and 35–40. (Students should work at least four or five from Exercises 7–14, in preparation for Section 1.5.)

1. Reduced echelon form: **a** and **c**. Echelon form: **b** and **d**. Not echelon: none.

3. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -5 & -10 & -15 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -5 & -10 & -15 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 0 & -1 & -2 \\ 0 & \textcircled{1} & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Pivot cols 1 and 2. $\begin{bmatrix} \textcircled{1} & 2 & 3 & 4 \\ 4 & \textcircled{5} & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 3 & 0 & -5 \\ 0 & 0 & \textcircled{1} & 3 \end{bmatrix}$

Corresponding system of equations: $\begin{aligned} \textcircled{x_1} + 3x_2 &= -5 \\ \textcircled{x_3} &= 3 \end{aligned}$

The basic variables (corresponding to the pivot positions) are x_1 and x_3. The remaining variable x_2 is free. Solve for the basic variables in terms of the free variable. The general solution is

$$\begin{cases} x_1 = -5 - 3x_2 \\ x_2 \text{ is free} \\ x_3 = 3 \end{cases}$$

9. $\begin{bmatrix} 0 & 1 & -6 & 5 \\ 1 & -2 & 7 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 7 & -4 \\ 0 & 1 & -6 & 5 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 0 & -5 & 6 \\ 0 & \textcircled{1} & -6 & 5 \end{bmatrix}$

Corresponding system: $\begin{aligned} \textcircled{x_1} - 5x_3 &= 6 \\ \textcircled{x_2} - 6x_3 &= 5 \end{aligned}$

Basic variables: x_1, x_2; free variable: x_3. General solution: $\begin{cases} x_1 = 6 + 5x_3 \\ x_2 = 5 + 6x_3 \\ x_3 \text{ is free} \end{cases}$

11. $\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & -4/3 & 2/3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Corresponding system:

$$\begin{aligned} \textcircled{x_1} - \frac{4}{3}x_2 + \frac{2}{3}x_3 &= 0 \\ 0 &= 0 \\ 0 &= 0 \end{aligned}$$

Basic variable: x_1; free variables x_2, x_3. General solution:

$$\begin{cases} x_1 = \frac{4}{3}x_2 - \frac{2}{3}x_3 \\ x_2 \text{ is free} \\ x_3 \text{ is free} \end{cases}$$

13. $\begin{bmatrix} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 0 & 0 & 9 & -6 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 0 & 0 & 0 & -3 & -3 \\ 0 & \textcircled{1} & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & \textcircled{1} & 9 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Corresponding system:

$$\begin{aligned} \textcircled{x_1} - 3x_5 &= -3 \\ \textcircled{x_2} - 4x_5 &= 1 \\ \textcircled{x_4} + 9x_5 &= -4 \\ 0 &= 0 \end{aligned}$$

Basic variables: x_1, x_2, x_4; free variables: x_3, x_5. General solution:

$$\begin{cases} x_1 = -3 + 3x_5 \\ x_2 = 1 + 4x_5 \\ x_3 \text{ is free} \\ x_4 = -4 - 9x_5 \\ x_5 \text{ is free} \end{cases}$$

15. The solutions found in Exercise 9 are $\begin{cases} x_1 = 6 + 5x_3 \\ x_2 = 5 + 6x_3 \\ x_3 \text{ is free} \end{cases}$. The equations corresponding to the matrix are

$\begin{aligned} x_2 - 6x_3 &= 5 \\ x_1 - 2x_2 + 7x_3 &= -4 \end{aligned}$. The result of substituting in for the basic variables is

$\begin{aligned} 5 + 6x_3 - 6x_3 &= 5 \\ 6 + 5x_3 - 2(5 + 6x_3) + 7x_3 &= -4 \end{aligned}$, which establishes that the solutions found to Exercise 9 are indeed correct.

17. The solutions found in Exercise 11 are $\begin{cases} x_1 = \frac{4}{3}x_2 - \frac{2}{3}x_3 \\ x_2 \text{ is free} \\ x_3 \text{ is free} \end{cases}$. The equations corresponding to the matrix

$\begin{aligned} 3x_1 - 4x_2 + 2x_3 &= 0 \\ -9x_1 + 12x_2 - 6x_3 &= 0 \\ -6x_1 + 8x_2 - 4x_3 &= 0 \end{aligned}$. The result of substituting in for the basic variables is

$\begin{aligned} 3((4/3)x_2 - (2/3)x_3) - 4x_2 + 2x_3 &= 0 \\ -9((4/3)x_2 - (2/3)x_3) + 12x_2 - 6x_3 &= 0 \\ -6((4/3)x_2 - (2/3)x_3) + 8x_2 - 4x_3 &= 0 \end{aligned}$, which establishes that the solutions found to Exercise 11 are indeed correct.

19. **a.** The system is consistent, with a unique solution.

b. The system is inconsistent. The rightmost column of the augmented matrix is a pivot column.

21. $\begin{bmatrix} 2 & 3 & h \\ 4 & 6 & 7 \end{bmatrix} \sim \begin{bmatrix} \textcircled{2} & 3 & h \\ 0 & 0 & 7-2h \end{bmatrix}$ The system has a solution only if $7 - 2h = 0$, that is, if $h = 7/2$.

35. Yes. The system is consistent because with three pivots, there must be a pivot in the third (bottom) row of the coefficient matrix. The reduced echelon form cannot contain a row of the form [0 0 0 0 0 1].

37. If the coefficient matrix has a pivot position in every row, then there is a pivot position in the bottom row, and there is no room for a pivot in the augmented column. So, the system is consistent, by Theorem 2.